

Raghav Gupta

📍 Zirakpur, Punjab, 140603 · 📞 +91 7973457736 · ✉️ raghavrgupta56@gmail.com · 🔗 linkedin.com/in/raghav-gupta-035b4a292 · 🐙 github.com/Raghav-56

About

Third-year Engineering student working in AI and DevOps. Fine-tuning LLMs & traditional models; Deploying web services & ML models; building RAG, ASR & API pipelines; and Automation scripts for various things. Fast learner, curious person.

Education

Bachelor of Engineering in Information Technology

2023 – 2027

University Institute of Engineering and Technology, Panjab University

- CGPA: 8.35/10 (till 4th semester)
- Reliance Foundation Undergraduate Scholar

Experience

AI/Research Intern | Design Innovation Centre, UIET

Jan 2025 – Oct 2025

- Developed computer vision pipelines using OpenCV, Ffmpeg, imgaug, etc. for keyframe extraction and video augmentation, supporting model training and making it more efficient.
- Built and optimized facial emotion recognition systems using Action Unit detection with pretrained DL models, improving inference accuracy.
- Researched and prototyped multimodal deep learning models (audio + vision) for enhanced emotion recognition accuracy.

AI/ MLOps Developer | Lafleurtech

Jun 2025 – Oct 2025

- Developed a local ASR multi-language transcription pipeline
- Tested and fine-tuned multiple models including Whisper, NVIDIA Parakeet
- Developed a system with speaker recognition and diarization, noise removal

Open Source Backend Intern | AISOC / PClub UIET

July 2025 – Aug 2025

- Engineered the model-hosting layer of PU-API, an internal API platform for consolidating deployment of DIC and university AI services with containerized inference endpoints.

Projects | Independent, Open Source & Hackathons

2020 – Present

- **Open Source (GitHub):** Active contributor with 500+ commits across ML tools, backend systems, automation scripts, with clean commit structure.
- **Hackathons – RAG Systems:** Designed and implemented various kinds of RAG system using FAISS, LangChain, and LLMs to extract insights from business documents, enabling automated decision support for industrial clients.
- **Cloud Deployment – Oracle:** Deployed team project on Oracle Cloud VM with network configuration, SSH access setup, and systemd service automation for zero-downtime reboot recovery.
- **Model Deployment – GCP:** Containerized and deployed ML model on Google Cloud Platform using Docker.

Technical Skills

Programming languages: Python, C++, C, Scripting (bash, ps1, python), JavaScript, TypeScript, LaTeX, Rust, Haskell, SQL, lisp/Racket, WASM

Libraries/Frameworks: PyTorch, TensorFlow, HuggingFace, OpenCV, Whisper, ffmpeg, NVIDIA NeMo, Scikit-learn, LLAMA factory

Dev Ops: Git, Docker, Oracle cloud, Google cloud platform, GitHub Actions, MLflow, Weights & Biases, Uvicorn, FastAPI

others: Working with CLI, linux OS, remote cloud servers(ssh config, key management), native UV (environment and dependency management), supermemory

Additional Information

Soft Skills: Teamwork, Technical Communication & Documentation, Problem-solving, Critical Thinking

Research Interests: Interpretability, Self-improving Agents, Multimodal Alignment, Agentic Workflows